



FIRST SEMESTER (2026-27)

In addition to part -I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : BIO G542
Course Title : Advanced Cell and Molecular Biology
Instructor-in-charge : VISHAL SAXENA
Instructor : Meghana Tare

1. Course Description:

The course deals with in-depth aspects of molecular biology of eukaryotic cell. We will discuss about molecules regulating and functional in Eukaryotic cell cycle: restriction point, G1 phase progression, role of cyclin and CKIs, cell cycle & cancer, metastasis, molecular chaperones, apoptosis and necrosis, Extra cellular matrix & adhesion, growth factors, cell signal transduction, stem cells and their potential in therapy, transport in and out of nucleus.

2. Scope and Objective of the Course:

This course will give insights to the students into some frontier areas in cell and molecular biology helping students understand the overall functioning of a cell under different environmental conditions including presence or absence of mitogens/ nutrition/ stress conditions, etc.. This will enable student draw concept map about different cell functions and related proteins.

3. Text Book (TB):

Molecular Biology of the Cell (7th/ 6th edition), Bruce Alberts, Alexander Johnson, Julian Lewis, David Morgan, Martin Raff, Keith Roberts and Peter Walter. Garland Publishing Inc., New York and London, 2022.

4. Reference Books (RB):

R1. Molecular Cell Biology (Ninth Edition), Lodish, Berk, Krieger, Scott, Bretscher, Ploegh, Matsudaira. Freeman, 2022.

5. Course Plan:

Module No.	Learning Objectives	Topics to be covered	Ref. Chaps.		Learning outcomes
			TB	RB	
1 (1)	Introduction	Course Orientation, discussion on mode of conductance of various evaluation components, seminar topics, etc			Planning of course conduct and operation
2-10 (9)	The cell cycle	Cell cycle Overview, different phases, cell cycle control and regulatory components, control of cell division and cell growth, Cyclin Kinase inhibitors	17	R1 20	Details about molecules functional in and regulating the cell cycle
11-15 (5)	Apoptosis or programmed cell death	Mechanism of cell death with special reference to apoptosis and necrosis, cascade of apoptosis implications	18	21, R1	How and why a cell dies? Molecular mechanism of Cell Death
16-18 (3)	Protein folding & Correction of misfolded protein	Mechanism of protein folding in <i>E.coli</i> and mammalian systems, Molecular chaperones, Protein trafficking across cell and export, Proteolysis	3 (3.2)	R1 13 (13.3)	Various molecules involved during native protein modification & breakdown





19-26 (8)	Mechanisms of cell communication	Mechanisms of signal transduction by cell surface receptor proteins (G proteins, enzyme coupled receptors), growth factors and their interaction with receptors in cell proliferation and its regulation, JAK – STAT, Wnt, Hedgehog, Notch, NF- κ B pathways.	15		Cell signaling is the basis of cross talk among cells. How does close cells and distant cells connect with each other will be learnt.
27-32 (6)	Cancer	Molecular basis of cancer, Cancer as micro-evolutionary process, cancer critical genes, molecular basis of cancer – cell behavior.	20	25, R1	Factors involved in causing cancer, their types and mis-regulations
33-35 (3)	Transport through nuclear pore complex	Mechanism for the transport of RNA and “Cargo” proteins into and out of nucleus	12	TB 12; R1 13 (13.6)	A well-organized structural specific mechanism of transport
36-41 (6)	Cell Junctions, cell adhesion and Extra cellular matrix	Cadherins, cell – cell adhesions, tight junctions and gap junctions, Role of CAMs in cell attachment, Basal Lamina, integrins, Components of ECM, role of ECM in bi-directional signaling.	19		The structure of cellular system.
42-43 (2)	Tissue renewal and Stem cell engineering	Epidermis and its renewal by Stem cells, multipotent stem cells, stem cell engineering	22		Details about stem cells and their functions.

Additional references will be provided as required.

6. Evaluation Scheme:

S.No.	Component	Duration	Wtg. (%)	Date & Time	Remarks
1	Mid-term test	90 minutes	30	10/10 AN2	Closed book
2	Quizzes & Assignment / Seminar/ Project	variable	30	To be announced	Closed Book/ Open Book
3	Comprehensive Examination	3 hrs.	40	16/12 AN	Partly Closed/ Open book

- 7. Chamber Consultation Hours:** To be announced in the class.
- 8. Grading Policy:** Award of grades would be guided by the histogram of marks and course average. **If a student obtains below 30% of the average of the top three performers or 40% of median (whichever is lower), or misses an entire evaluation component, ‘NC’ (Not Cleared) grade will be awarded.**
- 9. Notice:** Notices for tests will be displayed on Biological Sciences Notice Board.
- 10. Makeup Policy:** Makeups for quizzes, seminar or assignment will not be granted in any condition. Makeups for Tests will be granted only in case of severe medical urgency or hospitalization upon timely submission of valid proofs.

Instructor-in-charge
BIO G542

